CASE STUDY · FILMPAC

# Separating conflicting workloads with a hybrid data architecture

DATA ARCHITECTURE · MIGRATION · E-COMMERCE PERFORMANCE

| 3 | Zero | 10–15 ms |
|---|---|---|
| PURPOSE-SPECIFIC STORES | MIGRATION DOWNTIME | SEARCH-QUERY LATENCY |

## THE CHALLENGE

Filmpac's application data had to serve several very different needs: transactional application behavior, flexible relational reporting, fast partner data distribution, and customer-facing search and filtering. Treating those workloads as if they belonged in one datastore was creating friction in both development and operations.

The e-commerce path was especially visible. Search and filtering depended heavily on an overloaded WordPress/WooCommerce database, contributing to 5–20 second page loads and occasional waits approaching a minute.

## THE APPROACH

I redesigned the data layer so each technology had a deliberately narrow responsibility rather than allowing responsibilities to blur.

- PostgreSQL became the relational source for application data and flexible reporting.
- DynamoDB held canonical document representations used for partner distribution.
- Algolia served denormalized search and e-commerce indexes optimized for fast retrieval.

I added explicit adapter, validation, and transformation layers so changes moved between these representations intentionally rather than relying on fragile implicit synchronization. Multiple migrations were completed without production downtime.

Customer-facing search and filtering were moved from WooCommerce database queries to Algolia indexes, where search-query latency was approximately 10–15 ms.

## THE RESULT

The architecture improved reporting flexibility, partner data delivery, developer maintainability, and customer-facing performance while keeping data responsibilities understandable.

More importantly, the system no longer required one storage technology to be simultaneously optimized for incompatible workloads.

### WHY IT MATTERED

The value of a hybrid datastore is not the number of technologies. It is the clarity of their boundaries.

Each store had one job it was good at, while explicit transformation layers kept the system understandable and the data consistent.

## TECHNOLOGY

PostgreSQL · DynamoDB · Algolia · Zod · TypeScript · Data migration · Search indexing


ONE JOB PER STORE, AND NOTHING CROSSES BETWEEN THEM IMPLICITLY

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